



Customer

Engineering Solutions Ltd.

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Goods Receiver

Canadian Construction Corp

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Manufacturer

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Subcustomer

American Heavy Industries Inc.

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Digital Material Passport

ID	DMP-METAL-002	Version	1.0.0
Issue Date	2025-05-14	Certificate Type	EN 10204 3.1

Business Transaction

Order		Delivery	
Order ID	PO-78902	Delivery ID	DN-56790
Position	10	Position	1
Date	2025-04-21	Date	2025-05-13
Quantity	2000 kg	Quantity	2000 kg

Product Information

Product Name	Structural Steel S420N
Batch ID	H-10988-01
Heat Treatment	Normalized
Surface Condition	Hot-rolled
Production Date	2025-05-10
Country of Origin	DE

Product Norms

Designation	EN 10025-3 (2019)
Grade	S420N

Material Designations

System	EN
Designation	1.8902

Product Shape

Form	Plate
Length	6000 mm
Width	2000 mm
Thickness	25 mm

Chemical Analysis

Heat Number	H-10988
Melting Process	EAF+LF+VD
Casting Date	2025-05-09
Sample Location	Ladle

Elements

Symbol	C	Mn	Si	P	S	CEV
Unit	%	%	%	%	%	%
Min	-	-	-	-	-	-
Max	0.20	1.60	0.50	0.025	0.015	0.44
Actual	0.16	1.48	0.28	0.016	0.010	0.41

Formula Definitions

CEV = C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15

Mechanical Properties

Property	Symbol	Actual	Minimum	Maximum	Method	Status
Tensile Strength	Rm	560 MPa	520	680	EN ISO 6892-1	✓
Yield Strength	ReH	445 MPa	420		EN ISO 6892-1	✓
Elongation after fracture	A	24 %	19		EN ISO 6892-1	✓
Reduction of Area	Z	62 %	50		EN ISO 6892-1	✓
Charpy V-notch Impact - Energy	KV	58 J	40		EN ISO 148-1	✓
Brinell Hardness	HBW	185 HBW	150	220	EN ISO 6506-1	✓
Vickers Hardness	HV	195 HV10	160	230	EN ISO 6507-1	✓
Rockwell Hardness	HR	18 HRC		22	EN ISO 6508-1	✓
Elastic Modulus	E	210 GPa			EN ISO 6892-1	✓
Strain Hardening - Exponent	n	0.18			ASTM E646	✓
Plastic Strain Ratio	r	1.2	1.0		EN ISO 10113	✓
0.2% Proof Strength	Rp0.2	430 MPa	400		EN ISO 6892-1	✓

Validation

We hereby certify that the material described above has been manufactured and tested in accordance with the requirements of EN 10204:2004 type 3.1 and the specified standards. The results comply with the requirements.

Validated By

Name	Title	Department	Date
Johann Weber	Quality Inspector	Quality Assurance	2025-05-14